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Forward Kinematic Analysis of 6-R Industrial Robot Using Active Sampled Neural Network Regressor

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ABSTRACT

This paper describes the estimation of robotic forward kinematics using neural networks (NN). The quality of training data is essential for better accuracy of NN models. So, in this contribution, clustering-based refined training data selection is performed to improve the representativeness of the selected data. The k-means clustering algorithm is adopted to find the most distinct and informative training data. The 6-R MTAB Aristo-XT robot is selected as a case study to generate the experimental training and testing data for validation of ML techniques. Standard performance measures such as deviation error and mean square error (MSE) are evaluated, and graphical illustrations are presented for fair comparison of the results. Experimental results reveal that, instead of random sampling, the clustering-based active sampled training data selection is strongly suggested to improve the accuracy of NN regressor, and also it greatly reduces the time complexity to train the model.

KEYWORDS

6-DOF Robot, Artificial Neural Network (ANN), Clustering Algorithm, DH Parameters, End-Effector, Machine Learning (ML), Mean Square Error (MSE), Transformation Matrix

INTRODUCTION

Robotic kinematic models are associated with various mathematical functions to establish the relationship between joint angles and their respective positions. Generally, the joints are series-connected, which are controlled by the motors as actuators. Commonly, the series-connected links are attached with a gripper at the end. This defines the position of the robotic arm in the three-dimensional coordinate systems. The problem of estimation for robotic models can be widely categorized as 1) forward kinematics and 2) inverse kinematics. The equations derive relationships between the position of the gripper and joints, called forward kinematics (Saha, 2014). Similarly, estimating the joint angles from the end position information is termed inverse kinematics. Various Machine Learning (ML) approaches have been proposed in the past decade, addressing the generalization

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of robotic manipulators to predict the position of robots. The forward kinematics for a 6-3 Stewart Platform Mechanism (SPM) has been well established using the Neural Network (NN) (Yurt et al., 2007). Also, similar NN architecture helped to solve the forward kinematics problem for HEXA Parallel Robot (Dehghani et al., 2008). Further improvement is observed with integration of PSO to NN for a 6-DOF parallel robot (Li et al., 2007) to fine-tune the backpropagation learning. Also, an iterative approach to NN strategy proved to be efficient for the parallel manipulators (Parikh & Lam, 2009). Multi-Layer Perception (MLP) based backpropagation approach has been implemented for 3D cable robot (Ghasemi et al., 2010), 7-DOF Sawyer Robotic Arm (Theofanidis et al., 2018), and Delta parallel robot (Liu et al., 2019) model. Another hybrid approach combining neural networks with interval arithmetic to solve the forward kinematics (Schmidt et al., 2014) is experimented for Cable-Driven Parallel Robots.

Recently, the deep learning models for image processing have been vastly exploited and validated its significance on large-scale databases. Such a Hybrid Deep Learning approach is implemented for 6-DOF Serial Manipulator (Mohamed et al., 2020). The improved Newton iterative method is validated on a 3-DOF parallel manipulator (Wu & Xie, 2019) for the forward kinematics estimation problem. Other than machine learning, the optimization approaches are also found helpful to estimate the position information for 2-DOF using genetic programming (Arellano & Rivera, 2019). Also, the Artificial Neural Networks (ANN) for a 6-DOF robot is experimented with forward kinematics for different pose and joint space representations (Grassmann & Burgner-Kahrs, 2019). Inverse kinematics solution of a 3R robotic manipulator is also analyzed for optimal ANN parameters like input data, sample size, training algorithm (Karkalos et al., 2017) etc. The Path planning of industrial robot based on Fuzzy control is also investigated (Sahu & Choudhury, 2018).

From the above framework, it seems that the NN architectures are vastly exploited for non-linear estimation of the position while incorporation of optimal training sample selection is limited in the literature. All machine learning algorithms need a set of training dataset to train the model, which is determined prior. However, such an initial selection of data plays a crucial role in the prediction accuracy. Active selection of data samples is necessary, which may reduce the memory and time requirement of the learning process. Such an approach is based on a sparse dictionary (Patro et al., 2019), which eliminates the redundancy of selected training samples. Also, refining the training data for ANN (Kavzoglu, 2009), (Chen et al., 2013) proved to be efficient in terms of classification accuracy. Recently data sampling has been applied for deep learning architectures for image processing (Van et al., 2016). However, such analysis is limited in the forward kinematics prediction problems. Hence, presently, a clustering-based active training sample selection is proposed, which is fed to the NN model for learning and for further prediction of the gripper position.

FORWARD KINEMATICS MODELING

Multiple homogeneous transformation matrices are used to determine the location and orientation of the end-effector. With the information of joint position and rotational constraints, the forward kinematics problem may be well defined. The MTAB Aristo-XT robot's experimental setup is shown in Figure 1. An Aristo robot is a six axis articulated robot with all rotary joints.

For a 6-R robot, the transformation matrix between end-effector and base frame can be expressed as follows:

$$T_6 = A_1 * A_2 * A_3 * A_4 * A_5 * A_6 \tag{1}$$

Where, T_6 is a product of six homogeneous transformation matrices denoted by A_i (each $i = 1,2,3...6$ represents a joint). And each four-by-four transformation matrix A_i can be written as

Figure 1. Aristo xt 6-dof industrial robot in home position

$$A_i = \begin{bmatrix} \cos(\theta_i) & -\cos(\alpha_i)\sin(\theta_i) & \sin(\alpha_i)\sin(\theta_i) & a_i\cos(\theta_i) \\ \sin(\theta_i) & \cos(\alpha_i)\cos(\theta_i) & -\sin(\alpha_i)\cos(\theta_i) & a_i\sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

$$T = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{3}$$

Where, with reference to the base frame, the position of the end-effector is represented as vector p and the orientations are represented as n, o and a vectors. The a represents the approach vector, o represents the orientation vector and n is normal to both a and o vectors.

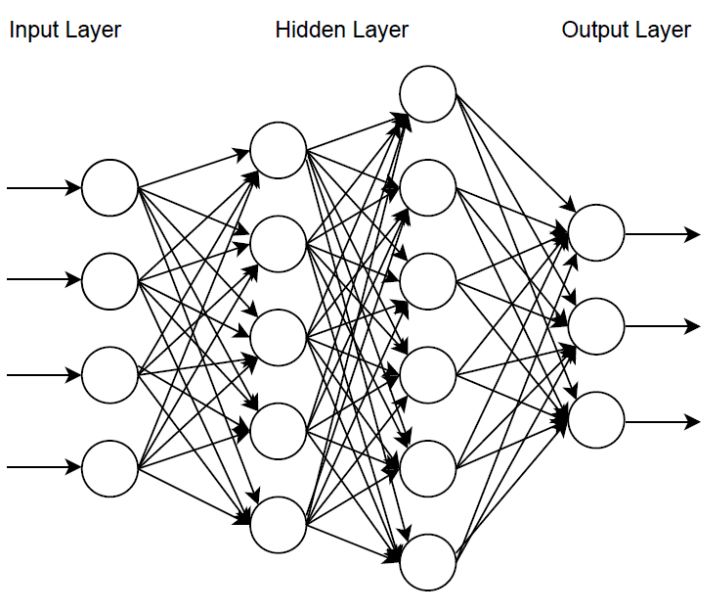
CLUSTER REPRESENTATIVE TRAINING SAMPLES FOR NEURAL NETWORK REGRESSOR

Neural Network Regressor

Artificial Neural Network (ANN) is a computational method for information processing that replicates the human brain with the use of mathematical models (Dora et al., 2020). ANN is made up of numerous layers with interconnected neurons (see Figure 2), each of which can be represented as shown in Figure 3. Independent variables characterize the input layer, while dependent variables characterize the output layer. The configuration of ANN for regression can be either 1) Multiple Input - Single Output or 2) Multiple Input - Multiple Output systems. The ANN can learn the relationship between input and output hyper-dimensional spaces based on the trained data. As a result, the trained model can be tested on real-time data to confirm its learning performance. The backpropagation strategy is commonly used in the ANN learning process, which updates the weight and bias of the neurons to minimize the error or loss function. ANN architectures are investigated and validated for a variety of applications, including classification (Panchal et al., 2016), regression (Eskandarian et al., 2017), pattern recognition (Abiodun et al., 2019), and so on. Instead of retaining all in-hand data (on an exponential scale), the informative and redundant data, which can be an efficient way to forecast the intended outcome, can be extracted. Many researchers have recently found this investigation to be fruitful, resulting in the use of ANN for a variety of prediction tasks. Several such predictions been proposed in the last decade for stock forecasting (Gao et al., 2017), electricity forecasting (Chen et al., 2017), labour productivity (Golnaraghi et al., 2019), and so on.

The general architecture of ANN is shown in Figure 2 comprising multiple layers. The input layers have neurons, exactly the same number of independent variables (feature space) in the dataset, while each neuron in the output layer refers to the prediction outcomes. Whereas, there may be multiple hidden layers and the size generally depends on the information in the input data, and are decided empirically with multiple trials. Furthermore, each hidden layer can contain several neurons, and researchers are still working to find out the exact solution.

Figure 2. General architecture of artificial neural network



A functional representation from the input (observation) space X to the output (decision) space Y can be expressed as an ANN (Eq. 4).

$$f : X \rightarrow Y \tag{4}$$

The linear relationship between the neuron's output Y' and its input feature space X with n features as $\left(x_1, x_2, .. x_n\right)$ can be stated as in Eq. 5.

$$Y' = \sum_{i=1}^{n} x_i w_i + b \tag{5}$$

Where, b is a bias term that is added to the sum of multiplicative weights $W\left(w1, w2, .., wn\right)$ with the input feature space.

In most of the real-time applications, the highly non-linear systems can be modeled with the help of ANN by introducing the transfer function. Such a transfer function is characterized by the activation function as shown in Figure 3 and Eq. 6.

$$Y = \varphi(Y') = \varphi\left(\sum_{i=1}^{n} x_i w_i + b\right) \tag{6}$$

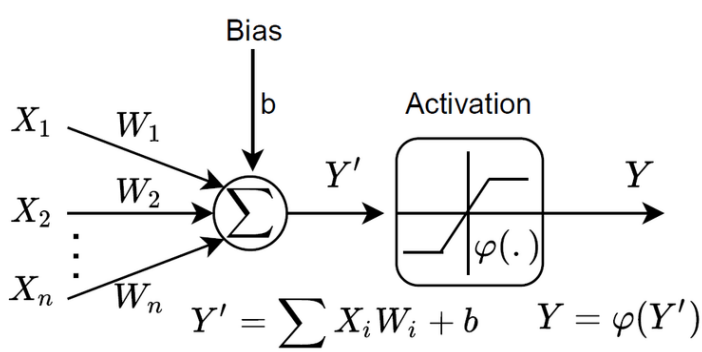
Where the activation function $\varphi\left(.\right)$ is often a nonlinear function such as $\log sig()$, $\tan sig()$, $purelin()$, and so on (Tan & Wong, 2017). The transfer functions are empirically chosen to best fit the given regression or classification task using a trial and error technique.

The ANN is generally trained using a backpropagation approach. The training is performed to reduce the error performance of the network by updating the weight and bias associated with each neuron. As usual, the random initialized weight and bias are updated with the help of a stochastic gradient descent algorithm to minimize the learning error.

PROPOSED CLUSTERING-BASED ACTIVE SAMPLE SELECTION

Any machine learning approach undergoes two stages: 1) training and 2) testing. Specifically, if the training is efficient, then testing performance can also be expected better. However, generally,

Figure 3. Building block of artificial neural network



the training data are selected randomly while evaluating the ML approaches. While, in the case of forward kinematic estimation problems, the data are collected experimentally and may have samples unevenly distributed over the three-dimensional work-space. Also, the collected data may be unevenly sampled at different sub-spaces. This may pose a problem for the ML approach to generalize the estimation over the entire work space. Hence, according to the available data, the training samples should be selected such that minimal training can lead to better performance of the ML approach. As presently the NN regressor is selected to solve the forward kinematics problem, its active training sample selection is explored further to improve its performance. The problem of training sample selection is obvious to obtain optimal performance of the regressor, and it is presently solved using an unsupervised clustering approach.

In this approach, K-means clustering is applied on position vectors $\left(p_x, p_y, p_z\right)$. The Number of clusters k for the K-means algorithm is selected with a prior assumption that the value of k should be the number of training samples. For M training samples, instead of random sampling, the clustering-based selection is proposed. Each cluster representative sample is selected for training the ANN. The output of the clustering approach is two dimensional matrix C of size $M\,X\,N$. Where M is the number of training samples and N is the feature space (in this case 3).

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ .. & .. & .. \\ C_{M1} & C_{M2} & C_{M3} \end{bmatrix}$$

Each element in the matrix C, M & N represents the cluster mean of the Mth cluster and N^{th} feature respectively. Currently, the features are considered as position vectors for $X, Y\ and\ Z$. location as $\left(p_x, p_y \text{and} p_z\right)$ respectively. Each vector $\left[C_{M1}\quad C_{M12}\quad C_{M3}\right]$ represents M^{th} the representative sample which has minimum distance to its own cluster center. According to the selected samples (position vectors) in the clustering process, their respective angle information are extracted and further training of ANN is carried out.

EXPERIMENTAL SETUP

To accomplish the forward kinematics analysis, the Aristo XT MTAB robot has been chosen. Table 1 lists the joint-link parameters for the 6-R MTAB Aristo-XT robot. The Aristo-XT robot has six

Table 1. Joint-link parameters (dh parameters) for the 6-r mtab aristo-xt robot

Joint_i	$\theta_i\left(\text{deg}\right)$	$\alpha_i\left(\text{deg}\right)$	$a_i\left(\text{mm}\right)$	$d_i\left(\text{mm}\right)$	θ_i^{min}	θ_i^{max}
1	0	90	0	158	-150	+150
2	90	0	300	0	+60	+120
3	160	90	0	0	+130	+190
4	-180	90	0	-378.5	-210	-150
5	0	90	0	0	-90	+90
6	0	0	0	64	-165	+165

degrees of freedom with all rotary joints. The first three degrees of freedom are located in the arm, which are used to determine the robot's position, and the next three degrees of freedom are located in the end effectors, which are used to provide orientation. The six joint angles (A_1 to A_6) within specified limitations are denoted as input variables, and the end-effector position world coordinates $X, Y \, and \, Z$ are expressed as output variables. The software used to simulate the robot is Aristo Version 1.4, as seen in the Figure 4. Palletizing, loading/unloading, glueing, spray painting, polishing, and object separation using vision systems are just some of the applications of the Aristo-XT robot.

Figure 5 shows the six angles in their various experimental variations. A total of 800 samples were collected and presented as the dataset. 20% of the data is chosen for training, with the remaining 80% being used for testing. In order to make a fair comparison between ANN and Active-ANN (A-ANN), an equal number of clusters are considered in the *k*-means clustering process i.e. total data is segmented into 160 clusters and for each cluster, their representative sample is selected for training and rest for testing purposes. For each $X, Y \, and \, Z$ location, the performances are evaluated in terms of MSE using tables, graphs as well as three-dimensional representations.

The proposed regression is simulated using the MATLAB18-b software, which is loaded on a PC with an Intel i-5 processor and 12 GB of RAM.

RESULTS AND DISCUSSION

Figure 6 shows a three-dimensional picture of the location estimation. In comparison to the ANN technique, the proposed A-ANN has better tracking capacity, as seen in the picture. The real trajectory is represented by the blue circled line, while the A-ANN and ANN trajectory of the robot manipulator are represented by the red dotted line and black dotted line, respectively. Figure 6 shows that when the

Figure 4. Coordinates of robots using aristo simulation

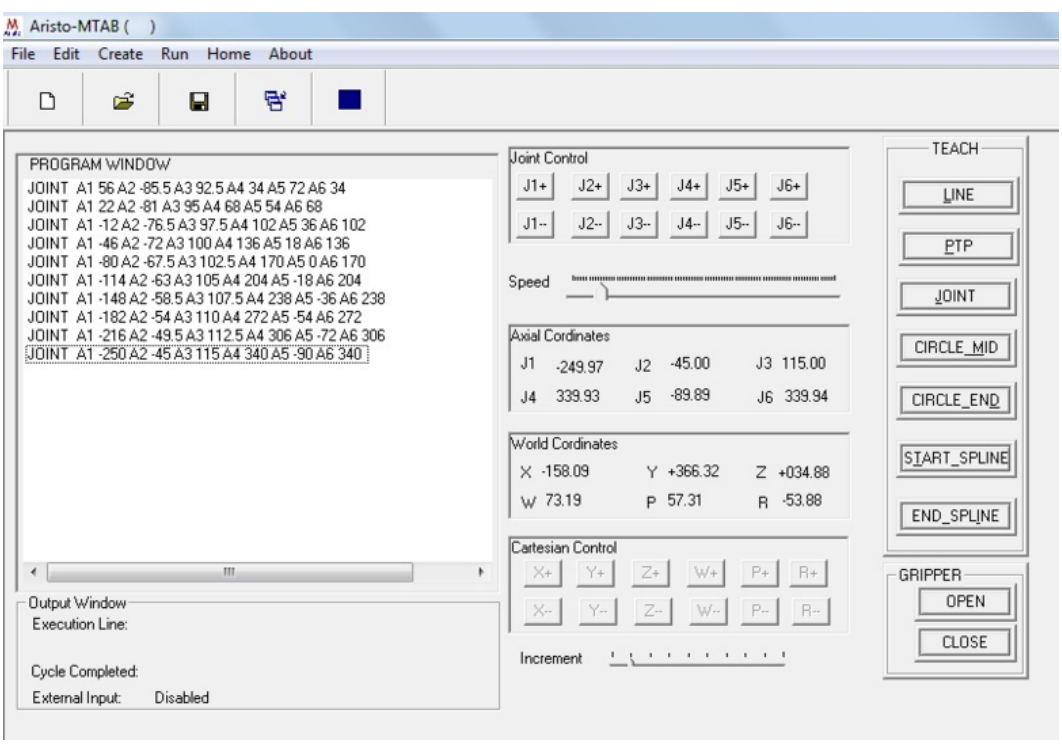


Figure 5. Experimental variation of angles for six rotary joints

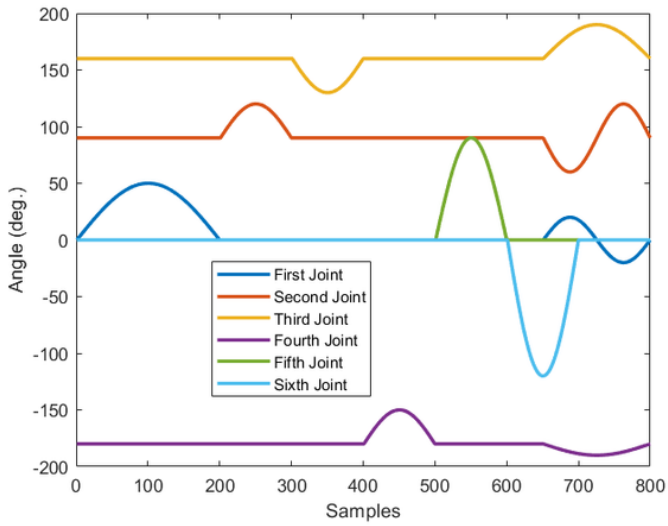
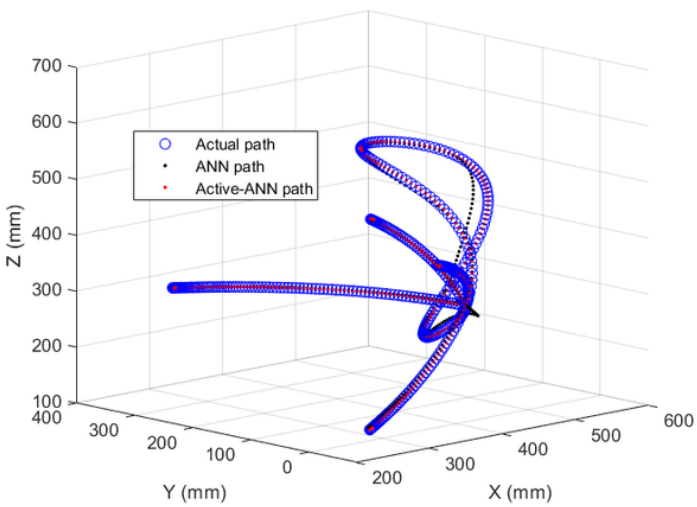


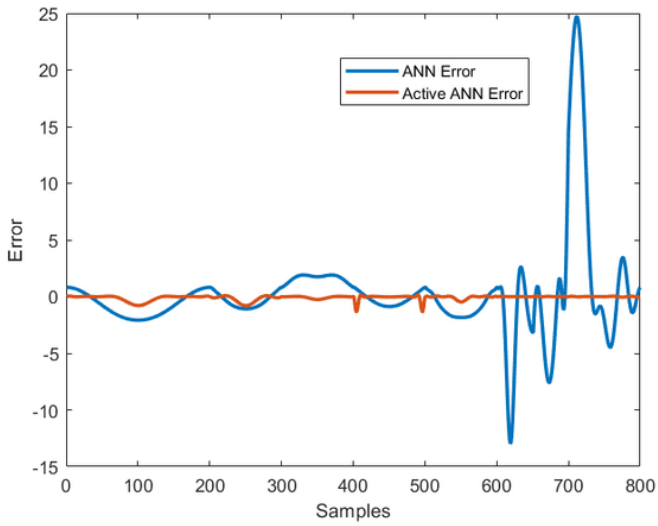
Figure 6. 3-d visualization of end effecter location, for experimental variation of six joint locations



proposed A-ANN approach is used, the resulting trajectory output closely matches the real trajectory. The ANN technique, on the other hand, does not follow the actual path and deviates from it at several points due to the presence of errors. A robot manipulator's tracking is to be precision and critical, especially when it's utilized in fields where speed is critical.

Figures 7, 8, and 9 show the error estimation for both the ANN and the proposed A-ANN for each X, Y, and Z location, respectively. In these figures, the ANN technique has large random variations in error, whereas the A-ANN approach has negligible variation in error. As a result, when compared to ANN, A-ANN requires less time to train the data.

Figure 7. X location error estimation for ann and active ann model



The MSE and Mean-MSE for ANN and the suggested A-ANN technique are shown in Table 2. The same number of samples are used in both the ANN and A-ANN studies. The MSE for the suggested A-ANN, on the other hand, is substantially lower than the ANN approach. From Table 2 it clearly shows that the mean MSE decreases from 5.31 to 0.24, which confirms the significance of the proposed A-ANN approach compared to the state-of-the art ANN technique. A similar interpretation can also be obtained from Figure 10's bar graph, which depicts the comparative error analysis of the tested models.

Figure 8. Y location error estimation for ann and active ann model

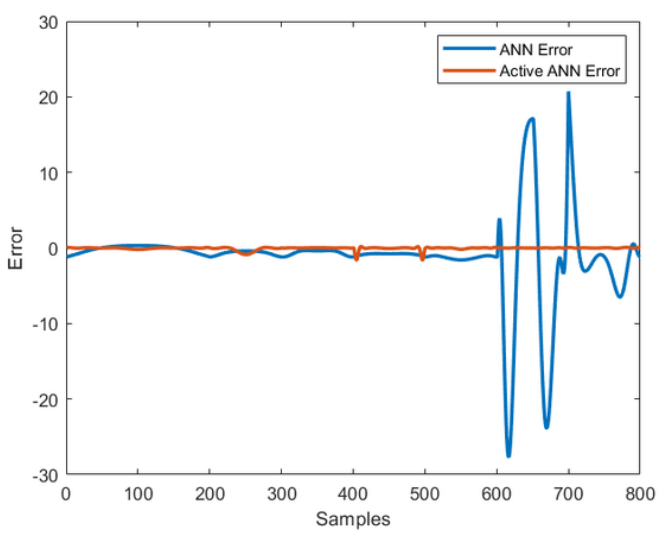


Figure 9. Z location error estimation for ann and active ann model

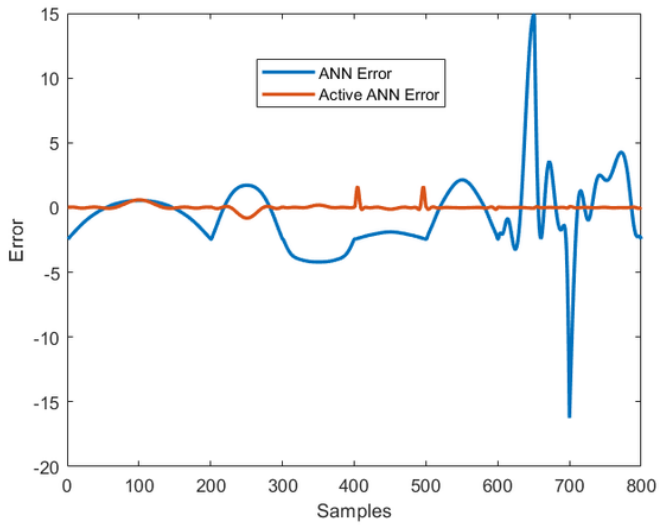


Figure 10. Bar plot indicating mse for ann and active ann model

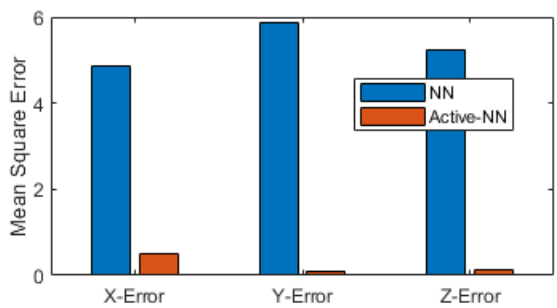


Table 2. Mean Square Error for X, Y and Z location with mean of MSE comparison for ANN and Active ANN model

$Joint_i$	X-MSE	Y-MSE	Z-MSE	Mean MSE	Samples
ANN	4.84	5.85	5.25	5.31	160
A-ANN	0.51	0.10	0.12	0.24	160

CONCLUSION

The non-linear forward kinematics problem is essential to solve using ml techniques to globalize the solution. Although the implementation of ann to solve the issue is exploited in the literature, further improvement is aimed at in the present study. The performance of the ann model totally depends on its training error. While, the training performance is fully dependent on the training data itself. So,

presently, the active selection of training data is exploited, experimented and validated. The selection is carried out with the help of a k -means clustering approach. Total data is divided into clusters, same as the number of training samples. For each cluster, the representative sample is considered training. This in turn selects the most distinct and informative samples which have substantially improved the regression accuracy. Also, for a specific in-hand mse for an arbitrary ann model, the number of samples for a-ann can be greatly reduced, signifying the complexity and time reduction in computation and ann learning. The obtained results strongly suggest the incorporation of clustering to select the optimal training samples, called active samples for ann models, to estimate the robotic forward kinematics problem. However, further experiments can be carried out to other robotic configurations for global validation of the proposed approach.

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